MIL-H-395F
28 January 1971
SUPERSEDING
MIL-H-395E
11 January 1966

MILITARY SPECIFICATION

HEATER, BITUMEN

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a trailer-mounted, self-contained, steam boiler with a three-tank-car heating capacity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

HH-B-671	- Brick, Refractory: Fire Clay.
HH-R-191	- Refractories; Fire-Clay, Plastic.
VV-F-800	- Fuel Oil, Diesel.
WW-P-406	- Pipe, Steel (Seamless and Welded) (for Ordinary Use).
WW-P-521	- Pipe Fittings, Flange Fittings, and Flange Steel and Malleable Iron (Threaded and Butt-Welding) 150 Pound.
ZZ-H-541	- Hose, Rubber, Steam.

Military

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).

FSC 3895

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MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military 5-Gallon.
MIL-B-3180	- Boilers and Related Equipment, Packaging of.
MIL-S-8660	- Silicone Compound.
MIL-E-11275	- Engines, Gasoline, Industrial Type, General Specification for.
MIL-C-13486	- Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty Single-Conductor and Multiconductor.
MIL-I-15126	- Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-I-22023	- Insulation Felt, Thermal and Sound Absorbing Felt, Fibrous Glass, Flexible.

STANDARDS

Military

MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1300	- Engine, Gasoline, Air-Cooled, 6 BHP, Military Design, Model 4A032, Installation Procedures.
MIL-STD-1400	- Engines, Gasoline or Diesel, Methods of Test.
MS17160	- Clip, Spring Tension (for Cylindrical Electronic Component).
MS21322	- Clamps, Loop - Aluminum - Plain and Cushioned.
MS21333	- Clamps, Loop - Steel - Plain and Cushioned.
MS21334	- Clamps, Loop - Double, Cable or Tube, Steel, Plain and Cushioned.
MS27142	- Connector, Plug, Electrical - Pin Contact No. 12, 14, & 16 A.W.G. Waterproof.
MS27143	- Connector, Plug, Electrical: Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical: Socket Contact, No. 14 & 16 A.W.G., Waterproof.

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MS27145 - Connector, Plug, Electrical: No. 14 A.W.G., Double Contact, Waterproof.
MS27146 - Connector, Receptacle, Electrical: Double Contact, Waterproof.
MS27147 - Adapter, Connector - 3 Way, No. 12, 14, & 16 A.W.G. Waterproof.
MS27148 - Contact; Pin, Electrical Connector: No. 12, 14, & 16 A.W.G.
MS33800 - Connector, Receptacle, Electrical - Pin Contact, No. 12, 14, & 16 A.W.G., Waterproof.
MS35387 - Reflector, Indicating, Clearance.
MS35388 - Tire, Pneumatic: Passenger Car, Light Truck, Truck and Bus.
MS35392 - Inner Tube: Pneumatic Tire.
MS35422 - Lampholder, Light, Marker, Clearance.
MS35423 - Light, Marker, Clearance - Service.
MS35424 - Light, Marker, Clearance - Blackout 24 Volt.
MS35643 - Gasket: Cap, Fuel Tank.
MS35644 - Filler Neck Assembly, Fuel Tank - Military Vehicles.
MS35645 - Cap, Fuel Tank: Tactical Vehicle, Fording.
MS35865 - Axle, Vehicular, Non-Driving, Single, for Tactical Wheeled Vehicles.
MS35876 - Generator, Engine Accessory, D. C. 28-Volt, 5-Ampere.
MS39020 - Band, Marker, Blank.
MS39062 - Circuit Breakers: Thermal, 30 Volt DC, Automatic, Waterproof, 15 to 30 Amp.
MS51330 - Stop Light - Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS51337 - Coupler, Drawbar, Ring - Heavy Duty.
MS53004 - Valve, Relay, Emergency-Breakaway, Air Brake.
MS53013 - Starter, Engine, Electrical, 24-Volt D.C., 3-Inch Frame Diameter, Solenoid-Actuated, Light-Duty.
MS53044 - Wheel, Pneumatic Tire - Disk Type, with Ring, for Tactical Wheeled Vehicles.
MS75020 - Connector, Plug, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.
MS75021 - Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.
Hs-39 - Construction and Industrial Machines Handbook Supplement.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N. Y. 10001.)

DEPARTMENT OF TRANSPORTATION (DOT)

Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the U. S. Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California, Department of Motor Vehicles, 2570 24th Street, Sacramento, California 95804.)

INDUSTRIAL PRESS

Machinery's Handbook.

(Application for copies should be addressed to the Industrial Press, 148 Lafayette Street, New York, N. Y. 14305.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Standards.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC. (TTMA)

RP#6 - Trailer Hitches and Connections.
RP#9 - Lighting Devices for Trailers and Containers.
RP#10 - Rear End Protection.
RP#12 - Trailer Air Brake System Test.
RP#14 - Heavy Duty Lighting System for Truck Trailers.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 1413 K Street NW, Washington, D. C. 20005.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

E-1 - Electrical Connector for Truck-Trailer Jumper Cable.
E-3 - Electrical Lighting and Wiring for Commercial Vehicle Bodies.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

AMERICAN WELDING SOCIETY (AWS)

Welding Handbook.
Standard Qualification Procedure, AWS B3.0.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI 100 - Prevailing-Torque Type Steel Hexagon Locknuts.
IFI 101 - Torque-Tension Requirements for Prevailing-Torque Type Steel Hexagon Locknuts.

(Application for copies should be addressed to Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, Ohio 44113.)

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FORGING INDUSTRY ASSOCIATION (FIA)

Forging Industry Handbook.

(Application for copies should be addressed to the Forging Industry Association, Illuminating Building, Cleveland, Ohio 44101.)

AMERICAN BOILER MANUFACTURERS ASSOCIATION

Standard Test Procedure for Packaged Firetube Boilers.

(Application for copies should be addressed to the American Boiler Manufacturers Association, 1180 Raymond Boulevard, Newark, N. J. 07102.)

DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be addressed to United States Department of Health, Education, and Welfare, National Air Pollution Control Administration, Publications Unit, 5600 Fisher Lane, Rockville, Md. 20852.)

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to Tire and Rim Association, Inc., Comand Building, 34 North Hawkins Avenue, Akron, Ohio 44313.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC.

Automotive Data Book.

(Application for copies should be addressed to the Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, N. Y. 10017.)

3. REQUIREMENTS

3.1 Description. The bitumen heater shall feature an oil-fired, horizontal steam boiler of the firetube multipass type with integral oil burner, fuel pump, gasoline engine, air blower, feed-water pump, combustion control, fuel-oil and water tanks, and all appurtenances necessary for a complete self-contained heater. A trailer chassis shall provide mobility for towing the

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heater by truck. The heater shall operate as specified herein with either one-, two-, or three-tank-car hookup, with the tank car(s) having a storage capacity of not less than 10,000 gallons each. Established operational component and vehicular ratings and capacities shall not be increased to meet the requirements of this specification. Except as specified herein, the bitumen heater automotive type components, assemblies and accessories shall be construed as "commercial automotive products", and they shall comply with all Federal Motor Vehicle Safety Standards applicable on the date of manufacture to nonmilitary, nonexport motor vehicle products.

3.1.1 Weights and dimensions. The bitumen heater gross vehicle weight (GVW) shall include the weights of the heater in operating condition with all attachments, equipment, and accessories, and to include full tanks and reservoirs. The physical characteristics of the heater shall be as specified in table I, when the heater is in travel position on level ground, and the inflation pressure of the tires is as recommended in the T&RA Yearbook and Supplementary Service Data Book for GVW operation at highway speeds.

Table I. Weights and Dimensions

Characteristics	Requirements
GVW rating, maximum	10,000
GVW distribution on towbar eye	3 to 8 percent of GVW
Overall length, maximum	192 inches
Overall width, maximum	96 inches
Overall height, maximum	96 inches
Ground clearance, excluding running gear, minimum	12 inches
Towbar eye height range (SAE J849)	31 thru 37 inches
Angle of departure, minimum (SAE J894)	25 degrees

3.1.2 Compatibility. The bitumen heater shall be compatible for towing behind a 5-ton, 6 x 6, M-51, military design dump truck.

3.2 First article.

3.2.1 Preproduction model. The supplier shall furnish a heater for examination and test within the time frame specified to prove, prior to starting production, that his production methods and choice of design detail will produce heaters that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Initial production. When specified, the supplier shall furnish to the Government one or more heaters for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. All materials shall be new and unused. Material not specified shall be selected by the supplier as being compatible with the intended use in accordance with the engineering and production standards of the heavy-duty boiler and pressure vessel industry and as set forth in publications of the American Society for Testing and Materials (ASTM), Society of Automotive Engineers (SAE) including SAE Handbook, Truck-Trailer Manufacturers Association, Inc., (TTMA), American Trucking Associations, Inc., (ATA), Tire and Rim Association, Inc. (T&RA), and Machinery's Handbook. Elastomer rubber products furnished shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the heater by the Government. Threaded parts shall be in accordance with the appropriate SAE standards. Maximum practical use shall be made of interchangeable hardware and a minimum number of types and sizes of bolts, capscrews, nuts, washers and similar common (standard) parts.

3.4.1 Sheet steel. Unless otherwise specified herein, sheet steel shall be SAE 1015, 1016, 1020, or 1025, hot rolled, condition and finish to suit the application.

3.4.2 Pipe. Unless otherwise specified herein, pipe shall conform to WW-P-406, weight A, class 1.

3.4.3 Pipe fittings. Unless otherwise specified herein, pipe fittings shall conform to WW-P-521, type I.

3.4.4 Fire-clay brick and tile. Fire-clay brick and tile shall conform to HH-B-671, class 5.

3.4.5 Plastic refractory. Plastic refractory shall conform to HH-R-191.

3.4.6 Insulation. Insulation shall conform to MIL-I-22023, type I, class 4.

3.5 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All belts, gears, shafts, pulleys, sprockets, spindles, drums, flywheels, chains, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. Exhaust or discharges

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from equipment shall be directed so that they do not endanger personnel. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. In addition, the heater shall be in accordance with the following:

(a) Conform to all requirements pertaining to large vehicles of the DOT Motor Carrier Safety Regulations, parts 390 and 393, except as specified herein.
(b) Conform to all requirements pertaining to lighting and braking of the California Vehicle Code, Division 12, except as specified herein.
(c) Conform to the Code of Federal Regulations, DOT Federal Motor Vehicle Safety Standards, except as specified herein.

3.6 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which are normally removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be located for accessibility.

3.7 Boiler. The boiler and accessories shall be built in accordance with the ASME Boiler and Pressure Vessel Code and shall be mounted and braced on the chassis. Each boiler shall have the capacity to convert continuously a minimum of 1,725 pounds of water per hour to steam from and at 212° F. The capacity shall be determined in accordance with the American Boiler Manufacturer's Association Standard Test Procedure for Packaged Firetube Boilers, except the working pressure shall be 135 psig. The boiler shall be of the firetube type and shall have a horizontal shell with front and rear heads of welded or bolted design to house a multipass construction. The combustion chamber shall be used as the first pass, and the remaining pass or passes shall be made by using standard boiler tubes.

3.7.1 Boiler shell. The boiler shell shall be a horizontal, cylindrical, steel pressure vessel of sufficient size for the capacity specified and to prevent large fluctuations in the water level and harmful carryover of wet steam at all pressures for which the boiler is required to operate. All welded seams shall be fusion welded and constructed in accordance with the ASME Boiler and Pressure Vessel Code for power boilers. The boiler shell shall be equipped with all necessary connections including steam outlet nozzle, feed nozzle, safety valve nozzle, and connections for the water column, blowoff, and injector. The shell shall be provided with handholds and inspection openings to permit inspection and access to all parts of

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the watersides for cleaning and shall have complete cover and yoke assemblies that shall be steamtight and watertight under all operating conditions specified herein. Bolted front and rear heads, when used, shall be sealed with asbestos gaskets.

3.7.2 Firetubes. Firetubes shall be of seamless steel or electrically resistant welded steel. The tubes shall be arranged so as to provide maximum rate of heat transfer and so as not to interfere with the natural circulation of the water in the shell or to prevent cleaning and flushing of the watersides. Repair, replacement, and cleaning of the tubes shall be from the front or rear of the boiler.

3.7.3 Refractories. The combustion chamber and rear head of the boiler shall be sealed with asbestos sheet or equivalent material between metal surfaces and refractory brick, tile, or clay. The rear head chamber shall have the capacity to permit the flow of hot gases to be reversed without undue deterioration. Fire clay or brick shall be used to line the rear head chamber. The combustion chamber shall be lined with suitable refractory material. When precast fire brick or tile is used, all joints shall be bonded with fire-brick cement. Plastic fire clay may be used in lieu of the fire brick.

3.7.4 Insulation. All exterior surfaces of the boiler except the doors, end covers, handholds, and areas providing heat to fuel-oil and water tanks shall be insulated with fibrous glass material as specified in 3.4.6. Insulating material shall be not less than 2 inches thick and shall be protected by a weathertight sheath constructed of No. 20 gage sheet steel. The insulating material shall be installed in such a manner as to prevent shifting during operation or travel as specified herein.

3.7.5 Boiler efficiency. The boiler fuel-to-steam efficiency shall be not less than 75 percent at the designated working pressure and at all capacity ratings from 30 to 100 percent. The efficiency shall be determined in accordance with the American Boiler Manufacturer's Association Standard Test Procedure for Packaged Firetube Boilers.

3.8 Boiler trim. Boiler fittings, blowdown valves, blowoff piping, feed-water piping, water column, pressure gages, injector, safety valves, main stop valve, check valves, fusible plug, a low water cutoff, and a high-pressure limit control shall be provided and shall comply with the applicable requirements of the ASME Boiler and Pressure Vessel Code. Provisions shall be made for feeding chemical into the boiler for internal water treatment. The water column shall include a column water gage,

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test valve, three trycocks, and blowdown valve. The water column and steam gage shall be located so that they are accessible and can be observed by the operator while the boiler is in operation. The steam-pressure gage shall be complete with siphon, gage cock, and inspector's test cock. The feed-water injector shall be complete with all necessary valves and piping for steam and water connections. Water connections shall be arranged so that the injector may supply feed water to the boiler from either the water tank on the heater or an outside source. Connections shall also be arranged so that the water may be taken from the outside source and discharged into the tank. An inlet strainer shall be furnished on the external intake pipe. The injector shall operate against steam pressures ranging from 45 to 150 psig. Each valve shall be identified by a permanent marking thereon or a corrosion-resistant metal identification plate fastened to each valve. The direction of operation shall be shown on each valve requiring manual operation or manual setting. The valve identifications shall be keyed to the operating instructions (see 3.30).

3.9 Oil burner. An oil burner shall be furnished and shall burn diesel fuel conforming to VV-F-800, grade DF-2. The fuel consumption of the burner shall be adjustable throughout the range of 4 to 17 gallons per hour. Tubing shall be copper or flexible, high-pressure, hydraulic hose.

3.10 Fuel pump. The fuel pump shall be of the positive-displacement type and shall be driven either direct or by belt from either the blower shaft or engine. The fuel pump shall have the capacity for pumping the fuel oil specified with a suction lift of not less than 15 feet and shall deliver the necessary quantity against the pressure required by the burner at full rated load plus not less than 25 percent for the oil circulation. An external bypass with a return line to the fuel-oil tank shall be provided.

3.11 Fuel-oil tank. A burner fuel-oil supply tank conforming to supplier's commercial standard except as specified herein shall be provided. The fuel-oil tank shall have a capacity of not less than 55 gallons. The tank shall be provided with a fuel tank cap gasket, filler neck, and filler-neck cap conforming to MS35643, MS35644, and MS35645, respectively. The filler-neck opening shall be located so that: (1) the tank can be filled from a 5-gallon can conforming to MIL-C-1283; and (2) spills and overflows shall not present a fire hazard. The following shall be stenciled near the filler-neck opening: "DIESEL FUEL ONLY". The tank shall also be provided with a drain, a fuel-level indicator, and a fuel-line valve. The fuel-tank outlet shall be located not less than 1/2 inch and not more than 1 inch above the bottom of the tank. The outlet fuel-line valve shall be located not more than 6 inches from the fuel tank. The tank shall be mounted to give protection from external damage and shall not present a fire hazard. All fuel lines shall be secured to prevent vibration and chafing and shall have no loops or dips that might permit the accumulation of water.

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3.12 Water tank. A water supply tank having a capacity of not less than 55 gallons shall be provided and shall be mounted to give protection from external damage. The water tank shall be made of hot-dip galvanized sheet steel having a minimum thickness of No. 14 gage. The tank shall be provided with connections to the injector and the water pump. A filler-neck cap and a vent shall be provided. The word "WATER" shall be stenciled near the filler opening.

3.13 Feed-water pump. An engine-driven, boiler-feed-water pump shall be provided for supplying feed water to the boiler from the feed-water tank or from a condensate return. The feed-water pump shall have all necessary fittings, valves, and connections, including strainer, check valve, and priming cup. The pump shall pump against a steam pressure of 150 psig. The feed-water pump minimum capacity shall be not less than the quantity required for continuous operation at maximum steam production.

3.14 Equipment base and housing. A metal base plate and a sheet metal housing shall be provided for the engine, blower, pumps, and burner assembly. The base plate shall be located on the heater so that each of these components are accessible for maintenance (see 3.6). A sheet metal housing shall be provided to enclose and protect these components when not in operation. The housing shall be of metal not less than a nominal 0.071-inch thickness and shall be reinforced. The housing shall be hinged so as to allow swinging into an open operating position. The housing shall be provided with hook type, tension latches, with strikers, that shall positively secure the housing in both the closed position and the operating position.

3.15 Power unit.

3.15.1 Military design engine. Unless otherwise specified herein, the heater shall be powered by a Government-furnished model 4A032 military design engine (see 6.4). The engine shall be mounted on and coupled to the heater in accordance with MIL-STD-1300. The engine shall run at 3,600 rpm for all heater operations specified herein. Speed reductions, as applicable shall be provided. The engine shall be provided with a 24-volt electrical cranking system to include starter, generator, voltage regulator battery and all required accessories to complete the cranking system. The starter and generator shall conform to MS53013 and MS35876, respectively, and shall be mounted in accordance with MIL-STD-1300. The following accessories shall be provided and installed in accordance with MIL-E-11275, group I: An instrument panel and instruments including an hourmeter, oil pressure gage and battery-generator indicator. The following remote controls shall be provided and installed with the control levers mounted on the instrument panel:

(1) Remote throttle control.
(2) Remote choke control.

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3.15.2 Commercial engine. When specified, the heater shall be powered by a commercial gasoline engine, including an electric cranking system. The electric cranking system shall include but not be limited to the following: An electric starter, generator, voltage regulator, battery and all other accessories required to complete the cranking system. The electrical circuit shall be 24-volt and shall have a negative ground. The following accessories shall be provided and installed: An instrument panel and instruments including an hourmeter, ammeter, and oil-pressure gage. The engines shall conform to the current Department of Health, Education and Welfare Standards for Exhaust Emissions, Fuel Evaporative Emissions and Smoke Emissions, Applicable to 1970 and Later Vehicles and Engines, Title 45, Part 85, as applicable.

3.15.2.1 Power and speed rating. The power and speed rating of the engine shall be such that it shall provide sufficient power and torque to operate the heater under any of the operational requirements and conditions specified herein, without exceeding the engine manufacturers published continuous horsepower rating, or 85 percent of the engine manufacturer's published maximum horsepower rating, whichever is lower.

3.15.2.2 Crankcase ventilation system. A positive crankcase ventilation system shall be provided on each engine. The system shall prevent or eliminate emission of crankcase vapors into the atmosphere under all operational conditions specified herein. If the ventilation system is connected to the air cleaner, it shall be connected in such a manner that no vapors or condensation will come in contact with, or be deposited on the air cleaner element. The design of the system shall be such that service or replacement of the system, or any part thereof may be accomplished without removal of other components of the engine.

3.15.3 Fuel tank. In lieu of the fuel tank specified in MIL-E-11275, a fuel tank conforming to supplier's commercial standard except as specified herein shall be provided. The tank shall be provided with a fuel shutoff valve attached directly to the tank, and a drain valve which is accessible. The fuel pickup point shall be not less than 1/2 inch above the bottom of the fuel tank. The tank and all lines shall be located or positioned to eliminate fire hazard caused by spilling, overflow, or drainage of fuel. The tank shall have a capacity of not less than 5 gallons. The filler neck, strainer, filler-neck cap, and filler-neck-cap gasket may be commercial. The filler-neck opening shall be not less than 3 inches in diameter and shall be located so that the tank can be filled from a 5-gallon can conforming to MIL-C-1283. The filler-neck cap shall be captive chained to the tank and shall incorporate an air vent and dust filter. A fuel-level indicator shall be provided. All fuel lines shall be secured to prevent vibration and chafing and shall have no loops or dips that might permit the accumulation of water.

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3.15.4 Clutch. A hand-operated or a self-energizing friction type clutch shall be furnished.

3.15.4.1 Clutch, overcenter engagement type. The heater shall perform as specified when a hand-operated, overcenter, engagement type clutch is furnished. The clutch shall transmit not less than 150 percent of the maximum torque developed by the engine.

3.15.4.2 Clutch, centrifugal. The heater shall perform as specified when a centrifugal self-energizing, friction type clutch is furnished. The clutch shall transmit not less than 150 percent of the maximum torque developed by the engine.

3.16 Mobility. The heater shall perform as specified when: (1) equipped with all attachments, equipment, and accessories stowed for travel and with all tanks filled to capacity; (2) towed over highways, secondary roads and rough unimproved roads specified, by a truck matched to the heater.

3.16.1 Highway travel. The heater shall travel over smooth, hard-surfaced, primary highways at sustained speeds up to 55 miles per hour (mph).

3.16.2 Secondary road travel. The heater shall travel over gravel and improved roads at speeds up to 30 mph.

3.16.3 Rough unimproved roads. The heater shall negotiate off-road terrain, ranging from rough unimproved roads, open rolling fields, through broken ground, and sand and gravel areas at speeds up to 10 mph. The trailer shall negotiate area grades up to 40 percent and traverse side slopes up to 20 percent. Without prior preparation, the heater shall ford hard-bottomed creek crossing up to 30 inches in water depth without damage or adverse effects due to immersion. After fording, the heater shall show no evidence of water leakage into brake, electrical, or other components required to be waterproof and shall not have an increase of water content in enclosures containing lubricants in excess of 2 percent of water volume, including land gearbox and wheel hubs.

3.16.4 Trailing and turning ability. The heater shall trail the towing vehicle without bottoming, cramping, weaving, or sidesway. The heater shall follow the truck without exceeding the tracking deviation, as specified in 393.70 of the DOT Motor Carrier Safety Regulations, when towed over a paved surface. Jackknife turns of at least 60 degrees, as measured from straight ahead to the right and to the left, shall not cause damage to or interference between the truck and the heater.

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3.16.5 Braking ability.

3.16.5.1 Service brakes. The combined service brakes of the truck and heater shall control, decelerate, and stop the vehicle combination, all in accordance with the California Vehicle Code and DOT regulations, when loaded at GVW as specified in table I. The brake system of the heater shall be in accordance with TTMA RP#12 except as specified herein. The air leakage within the heater's brake system shall not exceed 0.5 pound per square inch (psi) per minute with the brakes released, nor exceed 1.0 psi per minute with the brakes applied, each for periods of not less than 10 minutes duration.

3.16.5.2 Automatic actuation. The emergency breakaway system shall apply the heater brakes to stop and hold the heater when actuated by disconnect, breakage, or leakage of the emergency line.

3.16.5.3 Emergency-parking brakes. The emergency-parking brakes shall function to stop the loaded vehicle combination when the loss of approximately 50 percent of the main heater reservoir air pressure has occurred. In addition, the brakes shall provide a positive, continuous braking system, holding the wheels locked for parking of the heater on grades up to and including 30 percent, despite the depletion of the compressed air supply.

3.16.6 Suspension system. The suspension system shall withstand the imposed loads without evidence of overload, permanent deformation, wear, failure, or malfunction when the heater is subjected to the loading and the operating conditions specified in 3.1 through 3.1.2 and 3.16 through 3.16.4. Articulation shall provide not less than 7 inches difference in elevation, a minimum of 3 inches compression, and a minimum of 4 inches rebound, between opposite wheels on each axle without loss of tire-to-ground contact. For multiple axle suspensions, a minimum of 8 inches articulation shall also be required between all combinations of diagonal wheels without loss of tire-to-ground contact.

3.16.7 Coupling and jacking. The landing gear shall withstand, without damage or permanent deformation, the forces encountered in coupling and uncoupling the trailer from prime movers, maneuvering the trailer on a hard surface by personnel, and a free fall drop of not less than 6 inches onto a concrete surface. From its fully retracted position, the landing gear shall raise and lower the trailer with payload throughout its full range of travel, both with and without the support legs (see 3.17.4) in place under the rear of the trailer, without evidence of malfunction, binding, or interference.

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3.17 Trailer components. To provide the on-highway and cross-country mobility, the bitumen heater shall include a cart-type trailer as specified herein.

3.17.1 Intervehicular connections. The heater shall be furnished with an intervehicular electrical harness and two airbrake lines for connecting the truck electrical and airbrake systems to the heater trailer chassis systems. The airbrake service line shall be located on the curb side and the emergency line shall be on the road side, as standardized in SAE J702. All intervehicular connections shall be installed on the heater so that they extend forward between 36 and 42 inches toward the towing vehicle, and rearward at least 20 inches on the towbar, measured from the center of the towbar eye. A spring clip shall be installed to hold the electrical harness plug to the towbar with the harness doubled back on the heater. Two dummy couplings shall be installed to hold the airbrake line ends with the lines doubled back. Each dummy coupling shall be mounted perpendicular to the mounting surface and clearance shall be provided for the spring dust cover on the coupling. Intervehicular connections, when connected and when stowed, shall be routed or protected so that dropping of the tailgate of cargo trucks shall not cause physical damage.

3.17.2 Towbar components. A towbar eye conforming to SAE J847 shall be furnished and installed on the towbar within 1/8-inch tolerance of the longitudinal centerline of the heater. The towbar shall be within the height range specified in table I. An upper lunette stop, of a size of not less than 2 inches long by 1/2-inch square bar stock, shall be welded on three sides to the lunette support and shall function to prevent lunette rotations. Two safety chains, equipped with safety-type grabhooks, shall be furnished and shall conform to SAE J697 and to the DOT Motor Carrier Safety Regulations as specified herein. The forward length of each chain shall be as specified in 3.17.1 for intervehicular connections. The safety chain hooks shall pass through the shackles of the trucks. The heater attachment end of the safety chains shall be either perimeter welded or bolted to the towbar. The towbar shall be of the A-frame type. The towbar shall be constructed to embody structural strength to meet or exceed the following requirements: Support and transport the weight of the heater at GVW (see table I); transmit and distribute the imposed load to the suspension system and eye (see table I); and resist bending and twisting without permanent deformation or part failure when subjected to the performance requirements specified in 3.16 through 3.16.7.

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3.17.3 Landing gear. A single-leg landing gear, with capacity to fulfill the requirements specified herein, shall be installed on the towbar of the heater. The landing gear shall be vertically adjustable for not less than 12 inches. The landing gear shall either retract or be hinged and fold to provide not less than the ground clearance of the first trailer crossmember. The landing gear shall be arranged for positive locking into park and travel positions. A folding-type crank handle for operating the vertical adjustment mechanism shall be furnished. A travel retainer for the handle shall be provided. The tangential driver's effort at the handgrip to raise the maximum GVW load shall not exceed 30 pounds. The mechanism shall be protected from exposure to foreign matter which would impair functional efficiency. The landing gear shall incorporate a self-alining landing pad which will provide a ground bearing load not exceeding 20 pounds per square inch when the maximum towbar weight in table I is imposed at the towbar eye. The edges of the pad shall be turned up.

3.17.4 Support legs. Four adjustable support legs shall be furnished and shall be located at the corners of the heater. The height and adjustment shall provide leveling of the fully loaded heater, at any position, on slopes up to 18 percent. Each support leg shall be the retractable or the hinged and fold type that will provide the minimum ground clearance specified herein.

3.17.5 Frame. The frame shall be of the ladder type with longitudinal frame side rails reinforced with transverse crossmembers and gussets to meet the transport and the equipment operating conditions. The web of the side rails shall be continuous from the towbar to the last crossmember. Gusseted crossmembers shall tie together, maintain alinement, and distribute the stress of the suspension mounts.

3.17.6 Rear suspension. The heater shall be equipped with sprung, single or tandem axle suspension system, tailored to the heater. The suspension manufacturer's rated capacity shall be not less than the loading imposed on each member, measured at the ground, with the heater loaded to maximum actual GVW. The suspension shall provide a nondamaging ride cross-country and a safe, stable ride on-highway. The suspension shall have elastomer bushings or blocks that eliminate the need for lubrication. Tandem suspensions shall incorporate characteristics which shall provide freedom from brake hop. The installation of the suspension shall be in correct alinement with the frame and with the axle so that the two diagonal measurements from the center of the lunette to these suspension components do not differ in excess of 1/16-inch tolerance for every 20 feet or fraction thereof. Each axle shall also be parallel to the other axles, when tandems are furnished, so that the suspension wheelbases on each side do not differ in excess of 1/8-inch

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tolerance. Elastomer cushioned bumper stops or blocks shall be furnished to prevent axle-to-frame contact and limit suspension articulation at full compression (jounce) and rebound. Shock absorbers shall be furnished to control jounce and rebound.

3.17.7 Axle assemblies. Each single axle shall be rated by the axle manufacturer at a continuous load capacity rating of not less than 15,000 pounds for commercial highway trailer service. Each tandem axle shall be rated by the axle manufacturer at a continuous load capacity rating of not less than 10,000 pounds for the same service. The spindles shall be alloy steel forgings, correctly hardened and ground as recommended by the spindle manufacturer, and shall be located in the correct alinement position with the axle tubes. The spindles shall be equipped with assemblies of hubs, bearings, oil seals, and foundation brake of the various types as specified in 3.17.11.1. Hub assemblies shall be equipped with trailer industry standard tapered roller bearings having a standard inside-to-inside bearing spacing. Cups, bearings, and cones of all bearings shall be in matched sets from the same bearing manufacturer. The wheel bearings shall be oil lubricated. The viscosity of the oil shall be in accordance with oil lubricated system manufacturer's recommendations. The hub caps shall be provided with means for visual determination of the hub oil level. Provisions for replenishing the oil supply shall be incorporated. The oil seals and the system shall withstand the internal pressures and the external pressures incurred during all environmental and operational conditions specified herein, unless provisions for dust free and water free venting of each spindle have been incorporated. The axle hubs shall conform to SAE J694 and shall be of the 10-stud, 11-1/4-inch bolt circle type for conventional tires on single and tandem suspensions and of the 6-stud, 8-3/4-inch bolt circle type for tandem axle suspension with wide base tires.

3.17.8 Wheels, rims, and tires. The heater shall be equipped with single demountable disc wheel and tire assemblies. The disc wheels shall be of the heavy-duty type with T&RA rim and wheel disc constructed by the same wheel manufacturer and recommended for the service specified herein by the wheel manufacturer. Radial and lateral runout of the conventional tire wheels shall be within industry recommendations for over-the-highway commercial truck and bus wheels. Radial and lateral runout of the wide base tire wheels shall be as recommended by the wheel manufacture. The disc wheels shall have a standard, industry disc offset. Wide base tire wheels shall be limited to center or positive offset location of the disc. Rims, tires, inner tubes and flaps (if furnished), valve cores, and caps shall all be in accordance with the T&RA Yearbook and T&RA Supplementary Service Data Book. The tires shall be of not less than 100 level quality and shall be of the

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drive wheel type of tire. All tires shall be of the same size and load range. Commercial antiozonant compounds shall be included during tire manufacture. The load imposed on each tire, measured on level ground under loaded travel conditions, shall not exceed the recommended rated loads and inflation pressures of the tires for 50 mph highway speed for continuous highway operation. Valve caps shall be of metal and shall be equipped with elastomer inner seals. Valves shall be of the C-4, heat resistant, types.

3.17.8.1 Conventional tire assemblies. Conventional tires shall be installed unless otherwise specified herein. The heater shall be equipped with single tires of the tube or tubeless type, having a minimum size of 9.00 - 20, load range E (10 ply rating) or a minimum size of 10 - 22.5, load range E (10 ply rating). The tires shall have highway or traction lug tread.

3.17.8.2 Wide base tire assemblies. When specified, wide base, tubeless, single tires shall be installed. The single axle suspension shall be equipped with wide base tires size 16.5 - 22.5, load range H (16 ply rating), and tandem suspension shall be equipped with wide base tires, size 12 - 16.5, load range E (10 ply rating). The tires shall have traction lug tread.

3.17.8.3 Balancing. Tires, wheels, hubs, and brake drums shall be inherently in-balance, singly and collectively after assembly, to preclude wheel tramp, drumming, or excessive tire wear, when operated up to the maximum highway speeds specified herein.

3.17.9 Fenders and mudguards. The heater shall be equipped with two fenders and two rear splash guards. The fenders and splash guards shall be in accordance with SAE J321, SAE J682, and SAE J683. The fenders and splash guards shall restrict water and other road surface substances thrown from the heater wheels. The fenders either shall be contoured to the normal axle centers or shall be of octagonal shape about the wheels and shall include inner splash shields. The fenders shall support a weight of not less than 200 pounds imposed on any 1-square-foot area at any one time without damage or permanent deformation. The fenders shall be constructed of lightweight fiber glass reinforced plastic or aluminum material. The outer edges of the fenders shall be rolled to reduce side road splash. The splash guards shall be of a size not less than 12 by 18 inches, and shall include antisail characteristics. The guards shall be of fiber glass reinforced polyester or of solid polyvinyl chloride homopolymers and monomeric low temperature plasticisers without reinforcement. The physical properties of the guards shall have an ultimate strength of not less than 3,000 psi, a tear strength of not less than 375 pounds per inch, and a flex test life exceeding 1,000,000 cycles.

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3.17.10 Rear bumper. The heater shall be equipped with rear end protection conforming to DOT Motor Carrier Safety Regulations and to TTMA RP#10, except as specified herein. In addition, a horizontal rear bumper, extending the full width of the heater shall be required. The horizontal rear bumper shall be not less than 8 inches vertical height and shall extend rearward not less than 2 inches from any part of the heater. The left and right ends of the bumper shall be rounded and closed to prevent hooking when sideswiped by an obstruction. The rear end protection shall fold to the front during off-road transport, complying with both the angle of departure and the tire chain clearance specified herein.

3.17.11 Brake system. The heater shall be equipped with a water immersion resistant, straight airbrake system. The system shall conform to the DOT Motor Carrier Safety Regulations, the DOT Motor Vehicle Safety Standards, the California Vehicle Code Division 12, and the SAE Handbook, except as otherwise specified herein. All brake system components and plumbing shall be located, installed, and protected to prevent damage or malfunction from pallet and cargo movement and from ground obstructions, debris, and water and to obtain maximum safety, efficiency, and ease of maintenance. The brake system shall include service and emergency gladhands with dust covers, relay emergency valve, straight air cylinders or chambers, foundation brakes on each wheel, and all other components and plumbing required for a complete, matched brake system. The gladhands shall conform to SAE J318. The gladhands shall be colored blue for service and red for emergency. A spring-loaded, metal, dust cover shall be installed on each gladhand to prevent contaminants from entering the opening. The dust cover shall not cause interference with another gladhand with dust cover installed. The gladhands shall be installed on the heater as specified in 3.17.1. In addition, brakeline identification tags shall be furnished. The relay emergency valve shall conform to MS53004, except that the rapid bleed-down valve shall be replaced with a plug on the straight airbrakes. Conformance of the relay valve to the DOT regulations is mandatory. Each air reservoir shall conform to SAE J10. The service brake system shall have a total air storage capacity of not less than 2 times the minimum recommended air volume for truck-trailers in SAE J813. The reservoir shall be equipped with an automatic drain valve and a 90-degree twist drain cock. The drain cock shall be mounted on the low end of the reservoir and positioned on the curb side of the heater, accessible to personnel from the kneeling position.

3.17.11.1 Foundation brake assemblies. The total swept braking surface area of each single axle (two spindles) or of each tandem axle (four spindles) shall be not less than 510 square inches. The foundation brake

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components shall not be overstressed or exceed the component manufacturer's ratings under all operating conditions, including when a full, service brake application is made while the emergency-parking brakes are applied. An interlock system is acceptable. No part of the brake system shall extend below the bottom of the rim of the wheels. Self-adjusting slack adjusters for cam brakes or internal self-adjusters for wedge brakes shall be furnished. The self-adjusting slack adjusters shall be guarded or enclosed type providing protection from damage and frozen road slush. The self-adjusting mechanism shall function to maintain correct adjustment and compensate for lining wear and shall be matched for use with the higher heat capacity of metallic brake linings. The automatic brake adjusters shall fail safe in that, if failure occurs, means shall be provided to adjust the brakes manually. The straight air type foundation brake cylinders or chambers shall be of the spring applied or twin diaphragm mechanical lock type and shall include environmental protective covers and boots. Manual or slave air means to release the applied brakes shall be provided without requiring component or tools, other than furnished and stowed on each trailer. The brakes shall be equipped with removable dust shields and backing plates.

3.17.11.2 Brake drums and metallic linings. The brake drums shall be of the antibell-mouth type having SAE J431 automotive gray cast iron rubbing surfaces, SAE G3500 or SAE G4000. The drums shall be of the heavy weight type that will provide an adequate heat sink for use with metallic linings. The brake lining shall be nonfading from heat and from water immersion. The lining shall be of the metallic type and shall be coded class F E in accordance with SAE J866. The lining selected by the supplier shall be of a standard size as listed in the Automotive Data Book of the Friction Materials Standards Institute, Inc. The brake lining shall be installed on the shoes by the use of steel bolts and nuts, in the manner recommended by the lining manufacturer. The foundation brake wedges or camshafts, shoe rollers or piston pins, and anchor pins, as a minimum, shall be hard chrome plated or made of stainless steel that shall retard corrosion and prevent malfunction when the foundation brake assemblies have been exposed to salt water spray. All other ferrous surfaces of the foundation brake assemblies, between the dust covers and the drums, shall be protected by treatment conforming to MIL-T-704, type B, except that the finish coat shall not be required and that the metallic brake lining surface shall not be treated. The bearing surface shall be coated with a film of brake system silicone compound or high temperature grease lubricant.

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3.17.11.3 Brake lines. The brake lines shall be routed within the protection of the trailer framing members and shall be supported and secured to the trailer by elastomer cushioned, support clamps at intervals not greater than 3 feet. At each point where the lines pass through or near metal members, the lines shall be either protected by a grommet or loom and secured or a through-frame connector shall be used. The airbrake lines shall be of flexible hose with reusable fittings and shall conform to SAE J1402, types E or F. Not less than 5 inches of flexible air line slack shall be provided at each of the relay emergency valve connections to permit interchange of the various makes of MS53004 valves. Each brake line shall be identified adjacent to the valve connection. The routing and connection locations shall facilitate replacement of the brake lines from bulk supply.

3.17.12 Chock blocks. Four wood chock blocks and four cradles shall be furnished, two assemblies on each side. The chock blocks and captive chains shall be in accordance with figure 1 or may be of the commercial metal or plastic type. The chock block cradles shall be in accordance with figure 2 as modified to accept the chock blocks furnished. The captive chains shall be hooked onto the cradle's lower loop, with the excess chain figure-eight wound about the upper cradle ears. The lengths of the captive chains shall permit rapid removal of the chock blocks with uncoiling of the chain and facilitate placement of the chock blocks, both before and behind the trailer's forward suspension tires without unhooking the chain or entanglement of the chain. The cradle shall be installed on a vertical, flat surface with the ears up and the lower loop down.

3.18 Vehicular electrical system. The heater shall be equipped with a complete vehicular electrical system. Unless otherwise specified herein (see 3.19), the electrical system shall be 12-volt potential. Except as specified herein, the electrical system shall be in accordance with the following requirements:

(a) SAE Handbook, Electrical Equipment and Lighting Section.
(b) DOT Motor Carrier Safety Regulations.
(c) DOT Motor Vehicle Safety Standards.
(d) TTMA Recommended Practices RP#9 and RP#14.
(e) ATA Recommended Equipment Specifications E-1 and E-3.

All components shall be provided with the maximum practical protection from damage, shock, and vibration through means of location and resilient mounts. The heater component locations shall be especially selected to prevent damage or malfunction of the vehicular electrical system as a result of proximity and operation of the heater operational components.

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3.18.1 Wiring. The electrical connector receptacle with harness shall be as specified in ATA E-1 and E-3, and SAE J559 and located on the front as specified in 3.17.1. The leads from the receptacle shall be routed to a splashproof, trailer electrical nose box containing terminal strips and circuit breakers which shall protect all circuits of the trailer. Circuit breakers shall conform to SAE J553. All electrical wire and cable shall conform to SAE J555, SAE J558, SAE J559, and SAE J878 and shall have elastomer insulation. Thermoplastic insulation is not acceptable. Grommets shall be used to prevent chafing where wire or harness pass through any member. Cushioned support straps shall be furnished to prevent whipping, chafing, and other mechanical damage and to support wire cable and harnesses. Unsupported lengths shall not exceed 36 inches. Wiring shall be color coded in accordance with SAE J560. A complete wiring diagram plate shall be furnished in accordance with 3.30. Connectors and terminals shall be of the blade type complying with SAE J858, the pin type complying with SAE J928, or the spade type complying with SAE J561. Fully enclosed elastomer friction connectors shall be lubricated prior to assembly with electrical silicone compound. The size of wire shall be based on voltage drop and current requirements plus 15 percent increase to allow for deterioration. Wire splices shall be waterproof. Mechanical splicing devices shall have the interior waterproofed with silicone compound. Two or more wires shall be bound together in harnesses or protected within polyethylene trailer-type tubing.

3.18.2 Lighting. All running lamps and combination taillamps shall be of the waterproof type, having either elastomer housings with bulb and lens or elastomer ring mounts and sealed units. The light filaments shall be shock mounted to prolong life. Fourteen running lamps shall be furnished and shall be of the flush mount type, installed in industry standard 3-inch diameter holes. These lamps shall consist of four clearance lamps, four side-marker lamps, and six identification lamps, amber forward and red rearward. Four rear, SAE-SIT combination lamps shall be furnished. These lamps shall be of the flush or recess mount type having either a sealed unit or a lens and double contact, indexing base bulb. The combination lamps shall be installed behind guards at the rear of the body to prevent breakage when the trailer is backed into obstructions and to prevent obscuring of the lenses from spilled bitumen on the body. The forward area of these lamps shall be protected from road splash damage. All other lamps and reflectors shall be installed on the body and shall be recessed or guarded. No component shall be installed on the rub rails or extend beyond the maximum overall width of the body.

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3.19 Standard military trailer components. When specified, the heater shall be equipped with standard military components as specified herein. These requirements shall be in addition to the trailer components specified in 3.17 and the military components shall be furnished in lieu of their counterpart, commercial components, as applicable.

(a) The towbar eye, as specified in 3.17.2, shall conform to MS51337.
(b) The axle assembly, as specified in 3.17.7, shall conform to either the commercial axle or to MS35865-A military axle, except for straight air foundation brakes.
(c) Unless wide based tires are required (see 3.17.8), conventional 9.00 - 20, 8 ply rating, tire assemblies shall be furnished. Tires shall conform to MS35388-16, inner tubes shall conform to MS35392-13, and wheels shall conform to MS53044-5. A spare tire carrier and the spare wheel assembly shall not be required.
(d) Service and blackout electrical system. The heater shall be equipped with both service and blackout, vehicular, 24-volt, electrical systems having a common ground. The system shall be in accordance with figure 3, table II, and as specified herein. The service system shall be in accordance with 3.18, using the 24-volt components specified herein. The blackout system shall be similar to the service system, except that identification lights shall not be furnished.

The electrical components shall be in accordance with table II. Item quantities that are not specified shall be furnished as required (AR) for each trailer.

Table II. Vehicular Electrical System

Description	Requirements	Quantity per trailer
Service lamps; clearance side-marker, and identification	MS35423	14 min.
Blackout lamps; clearance and side-marker	MS35424	8
Running lamp installation	Horizontal, using information on MS35422	AR
Service combination lamps	Commercial SAE-SIT, 4-in. dia., waterproof, double contact (SAE J759), 24-volt	2

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Table II. Vehicular Electrical System (cont'd)

Description	Requirements	Quantity per trailer
Blackout combination lamps	MS51330 (military type)	2
Reflectors	MS35387	8
Front trailer receptacle	MS75021-1 with matching 4-bolt spring cover	1
Trailer nose box or two junction boxes	Figure 3, waterproof type	1 Assy.
Low tension cable	MIL-C-13486, 14 AWG min.	AR
Intervehicular harness	MIL-C-13486/12 with MS75020 plugs on each end, matching truck and heater (see 3.17.1)	1 Assy.
Insulation	MIL-I-15126 and MIL-I-631	AR
Loom	Polyethylene trailer tubing or SAE J562, oilproof loom	AR
Cushioned support clamps	MS21322, MS21333, and MS21334	AR
Circuit number bands	MS39020	AR
Waterproof friction connectors	MS27142 thru MS27148 and MS33800	AR
Connector and splice lubricant	MIL-S-8660	AR
Connector holder clip	Plastic or metal spring clip similar to MS17160-29	1 per friction connector
Threaded plugs and receptacles	Waterproof military type	AR
Terminal connectors	Spade type per SAE J561	AR
Terminal sleeves	Heat shrinkable tubing	AR
Circuit breakers	MS39062	3

3.20 Lubrication system. All moving components requiring periodic lubrication shall be provided with a means for lubrication. Lubricant types shall be selected in accordance with SAE J754, except as specified in 3.17.7. Components requiring grease lubrication shall be provided with accessible, SAE J534 pressure grease fittings with surface check balls. A lubrication chart plate in accordance with SAE J753 shall be installed on each trailer. Plate material and method of fastening shall be as specified in 3.30 for the data plates.

3.21 Servicing and adjustment. Prior to acceptance by the Government, the supplier shall service and adjust each trailer for operational use including, but not limited to, the following: Adjustment of electrical and

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brake systems, torquing of the wheel lug nuts, inflation of all tires, and complete lubrication with grades of lubricant recommended for the ambient air temperature at the delivery point. Adjustment of the service brakes shall prevent the shoes from dragging on the drums with the brakes released and shall provide the clearance recommended by the brake assembly manufacturer for new brake shoe metallic lining. Adjustment of brake assemblies, having slack adjusters and camshafts, shall provide a 90-degree angle between the push rod and the centerline of the slack adjuster when the push rod is at one-half of the available travel.

3.22 Accessories. The heater shall be provided with eight 25-foot and four 12-1/2-foot lengths of 1-inch, high-pressure, braided, galvanized-wire-reinforced, rubber steam hose conforming to ZZ-H-541, type I, grade A. Hoses shall be provided with couplings in accordance with ZZ-H-541. A steam-cleaning-jet attachment shall be furnished with connections for the steam hose supplied. A gate valve shall be provided on the cleaning attachment. The steam-cleaning-jet attachment shall be furnished with heat-insulated handholds. Provisions shall be made to carry the accessories on the heater when they are not in use. Accessories shall remain in their carrying places when the heater is transported and operated as specified herein.

3.23 Workmanship. Defective parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished without approval of the contracting officer. All workmanship on the heater shall be in accordance with the engineering and production standards of the bitumen heater and heavy trailer industry. The heater shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the heater before and after painting shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.23.1 Metal welding. All metal welding shall be in accordance with the American Welding Society's Welding Handbook and as specified herein. Welds shall provide the maximum notch toughness and fatigue strength with a minimum of residual stresses in accordance with the Welding Handbook. All welds made on high-stress members shall be made from core or heavy-coated electrodes, gas-shielded, or submerged arc. Electroslag and electrogas processes shall be included for use on heavy sections, 1 inch or more in thickness. Plates and shapes having thicknesses of 3/4 inch or less shall require welds of 100 percent penetration for butt welds and not less than 67 percent penetration for fillet welds. Welds which directly transverse or which induce fatigue in high-stress members, and

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spot, tack, or intermittent welds for strength are not acceptable. Deficiencies in workmanship shall not be covered up by using filler materials. Fillet welds shall be provided, when necessary, to reduce stress concentrations. The surfaces to be welded shall be free from scale, paint, grease, and other foreign matter. When multiple layers of weld metal are required, each layer, or each layer after the third pass with multipass welding process, shall be thoroughly cleaned before depositing another layer. All welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the part or assembly. After welding, all weldments shall be free from slag (except mill scale), flux, weld spatter, and other factors which affect the quality of the weld. Except for vertical welding of the electroslag process, the work shall be positioned for flat welding whenever practicable. No undercuts or overlays will be permitted on any weldment. Butt-type joints shall be alined within 20, 10, 7.5, and 5 percent of the thickness of the equal or thinnest members involved, for metal thicknesses of 1/32, 1/4, 1/2, and 1 inch minimums, respectively. Where different metals are welded to each other, filler metal having strength of the lower strength material shall be used. When the same metals, or metals having similar properties are used, the deposited filler metal shall develop the full design strength of the particular weld joint or weldment. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification. Weld joints exceeding 50,000 psi shall be inspected for soundness. Prior to production fabrication of any weldment of these joints, the supplier shall determine the inspection medium (magnetic particle, radiographic, dye penetrant, ultrasonic, gamma ray, or eddy current) necessary to determine the soundness of production welds.

3.23.2 Aluminum welding. Aluminum welding shall be accomplished by the gas-metal-arc or the gas-tungsten-arc method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations. When machines are available, all aluminum welds shall be made by automatic machines. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed or chipped; hammered welds shall not be acceptable. Preheating for welding is permissible, providing the temperature does not exceed 400° F. for a total time of 30 minutes. Aluminum weld processes requiring the use of welding flux shall not be used.

3.23.3 Brazing. Brazing of aluminum and aluminum alloys shall be performed in accordance with nationally recognized and accepted recommended procedures and standards of the aluminum manufacturing industry.

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3.23.4 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have burrs removed. Integral or separate washers or lockwashers shall be provided. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All capscrews, bolts, and nuts shall be tight and correctly torqued.

3.23.5 Critical bolted connections. Critical load carrying bolted connections shall, in addition, be as specified herein. Bolt holes shall be reamed as required. All critical junctures and joints shall be fastened with high strength bolts with washers and nuts. Bolts shall be not less than SAE J429, grade 5, and shall be of higher grade as necessary. Washers shall be of the flat hardened steel type, sized to spread the clamping loads such that brinnelling or embedding in the component members shall be prevented. Mating nuts shall be of the self-locking type in accordance with IFI Standards 100 and 101. Fasteners shall be lubricated prior to wrenching, and installation tools shall be proofed at set intervals to insure constant and correct torque performance.

3.23.6 Riveted connections. Riveted connections shall be in accordance with SAE J492. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.7 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended use.

3.23.8 Forgings. Forged components shall be in accordance with the Forging Industry Association Handbook.

3.23.9 Machine work. All machined parts shall be manufactured to gage through the use of correct jigs and fixtures and shall permit interchangeability of these parts.

3.24 Electromagnetic interference. Unless otherwise specified herein, the electromagnetic interference of the heater shall be in accordance with MIL-STD-461, III D.

3.25 Fungus and moisture resistance. Unless otherwise specified, the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

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(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.26 Lifting attachments. Permanently affixed attachments that enable the heater to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the heater, 2-1/2 times the resultant vectorial force imposed when the heater is suspended by the lifting attachments. Attachments for multiple slings shall be located so that: (a) slings will converge at not more than 19 feet above the lowest extremity of the heater, and (b) not less than 1-inch clearance will be maintained between the slings and the heater. The inside diameter of the attachment eye shall be not less than 3 inches. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.27.

3.27 Tiedown attachments. Permanently affixed attachments that permit fastening the heater to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand, without damage to the heater, its proportion of the following static loads applied to the heater: 8-G forward, 8-G rearward, 2-1/4-G upward, 1-1/2-G sideward, where G equals cargo weight of the heater. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.26.

3.28 Identification marking. The heater shall be identified in accordance with MIL-STD-130. The marking shall be applied to the heater on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.29 Shipping data plate. Shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in addition, shall show the silhouette of the heater in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.30 Instruction plates. Each heater shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the heater. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.31 Treatment and painting. The portions of the heater normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.32 Vehicle marking. The registration numbers and other marking of the heater shall conform to MIL-STD-642. The GVW shall be marked on each forward side of the heater in a manner to be discernible. The prescribed tire pressure shall be marked on the heater near the tires. "Lift" and "Tiedown", as applicable, shall be marked adjacent to their respective lifting and tiedown attachments. Marking shall be not less than 1 inch high.

3.33 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.34 Tool space. A tool space shall be provided below the boiler at the towing end of the heater. The tool space shall have a hinged access door of sheet steel with a minimum thickness of No. 14 gage. The hinged door shall open not less than 90 degrees and shall have a vibrationproof latch. The tool space shall be large enough to contain the tools specified herein.

3.35 Items to be furnished with heaters. No shipment of heaters shall be made unless repair parts, maintenance tools, technical publications, and accessories are included with the shipment or unless approval for shipment of the heaters without such items has been received from the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The supplier is responsible r insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Engine performance. Prior to examination and tests of the heater that is powered by a commercial engine, the engine that will be installed in the preproduction model unit shall be tested in accordance with test method 2300 of MIL-STD-1400. Nonconformance to 3.15.21 or failure to meet the requirements of test method 2300 of MIL-STD-1400 shall constitute failure of this test.

4.3.2 Examination. The preproduction heater shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests. The preproduction heater shall be tested as specified in .6.2 through 4.6.2.10. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. An initial production heater will be selected at random by the Government from the first five heaters being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 through 4.6.2.10 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the initial production heater shall not exclude the remaining heaters from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of the initial production heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production heater and shall be cause for refusal by the Government to continue acceptance of production heaters until evidence has been provided by

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the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on heaters previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all heaters accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each heater shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each heater shall be tested as specified in 4.6.2.11. Failure of the test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The heater shall be examined for the following defects

101. Assembly not complete or not correct.
102. Parts or components missing or do not function as specified.
103. Ease of maintenance not as specified.
104. Material not as specified.
105. Safety devices not as specified.
106. Boiler not as specified.
107. Boiler trim not as specified.
108. Oil burner not as specified.
109. Fuel pump not as specified.
110. Fuel oil tank not as specified.
111. Feed-water pump not as specified.
112. Water tank not as specified.
113. Gasoline engine or engine accessories not as specified.
114. Chassis not as specified.
115. Accessories not as specified.
116. Lubrication not as specified.
117. Fungus and moisture resistant not as specified.
118. Lifting or tiedown attachments not as specified.
119. Identification marking, shipping data, identification plates, or instruction and warning plates missing or not as specified.
120. Treatment and painting not as specified.
121. Stenciling not as specified.
122. Repair parts and tools missing or not as specified.
123. Tool space not as specified.
124. Workmanship not as specified.

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4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The heater shall operate as specified herein without maintenance other than the supplier's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the supplier prior to test.

4.6.2.2 Boiler construction. Test the boiler and report the data on the required format in accordance with the latest revision of the ASME Boiler and Pressure Vessel Code for the specified pressure and applicable requirements for boiler appurtenances. Failure of the boiler or any appurtenance to meet the requirements of any portion of the code or failure of the boiler to be stamped after the test with the ASME symbol shall constitute failure of this test.

4.6.2.3 Performance. Operate the boiler at a steam working pressure of 135 psig for 2 hours. Determine and report the boiler capacity rating and thermal efficiency of the boiler in accordance with the American Boiler Manufacturer's Association Standard Test Procedure for Packaged Firetube Boiler. Determine the fuel consumption rate of the oil burner and the fuel pump capacity. Observe the operation of the feed-water injector and the feed-water pump. Inability of the boiler to convert continuously a minimum of 1,725 pounds of water per hour to steam from and at 212° F., a boiler fuel-to-steam efficiency of less than 75 percent, inability of the oil burner to burn diesel fuel conforming to VV-F-800, DF-2 in accordance with 3.9, inability of the fuel pump to conform to 3.10, inability of the feed-water injector to conform to 3.8, or inability of the feed-water pump to conform to 3.13 shall constitute failure of this test.

4.6.2.4 Road test. The following road test shall be performed. Couple the heater to the towing vehicle specified herein and tow the heater as specified in table III, with a full complement of fuel and water in the storage tanks.

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Table III. Road Test

Course	Mileage and Speed	Loading
Primary Highway	200 miles at variable speeds up to 55 mph.	Loaded
Secondary Highway	50 miles at variable speeds up to 30 mph.	Loaded
Cross-country	50 miles at variable speeds up to 10 mph.	Loaded

During the road test, perform and record the actual results of the following:

(a) Tow the trailer on side slopes up to 10 percent, having the curb side downhill and having the road side downhill. Make full-throttle starts and panic stops on the side slopes.

(b) Tow the trailer over 7-inch-high obstacles such that all combinations of the wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or interference, and, after returning the suspension to normal, measure for evidence of misalinement.

(c) Tow the trailer over V-ditches not less than 36 inches wide and 18 inches deep. Observe ground contact areas for possible heater permanent deformation other than paint. Negotiate the ditches both perpendicular and at an angle of 45 degrees to the direction of travel, until at least 10 V-ditch crossings have been completed.

(d) Measure and record the deviation or off tracking of trailer with respect to the prime mover, when the trailer is towed straight ahead on a paved surface.

(e) Jackknife the trailer at angles up to 60 degrees from straight ahead both to the left and to the right.

(f) Turn the prime mover, towing the trailer, in a rolling, sharpest-turning circle for two complete turns and repeat in the opposite direction.

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(g) Turn the trailer lights on, and immerse or back the trailer into water not less than 20 inches deep for not less than 2 minutes. Stop the trailer using the trailer brakes and turn off the lights for not less than 10 minutes. Turn on the lights. Tow the trailer out of the water and make a brake stop using the trailer brakes and observe trailer braking for water fade, grabbing, and chatter. Observe all brake system components and all electrical system components, except hermetically sealed lights, and inspect at least the hubs and bearings of one wheel for damage and test the lubricant for water content increase.

Failure to complete the test, inability to operate and function as specified herein, or nonconformance to 3.16 through 3.16.7 shall constitute failure of this test.

4.6.2.5 Brake test. Test the trailer for conformance to DOT Motor Carrier Regulations, the California Vehicle Code, and to the requirements specified herein. Record all data.

(a) Tow the trailer over the test course and when the trailer reaches a speed of not less than 20 mph, make a panic brake stop using both prime mover and trailer brakes. Perform the test four times, allowing not more than 3 minutes between tests.

(b) Tow the trailer at not less than 50 mph and apply service brakes to produce a deceleration rate of 10 feet per second, until the trailer has slowed to 30 mph. Then immediately accelerate at wide-open-throttle again up to not less than 50 mph. Repeat the snubbing cycle for a total of 5 cycles and make a panic stop on the 6th cycle.

(c) Tow the trailer over the test course and, when the trailer reaches 20 mph, cause the trailer to go into "emergency", simulating a trailer breakaway from the prime mover. Restore the trailer brake system to normal. Disconnect the trailer emergency line hose and observe if there is an air leak at the trailer gladhand.

(d) With the trailer stopped on a paved surface, uncouple the prime mover brake lines from the trailer and when straight airbrakes are furnished, bleed down all trailer

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air reservoirs. Determine by the drawbar pull method or on actual grades the maximum holding ability of the trailer emergency-parking brakes. Release the emergency-parking brakes using only tools supplied with each trailer or using a slave air source. Insure correct and rapid emplacement of the chock blocks.

(e) Test the trailer brake system as specified in TTMA RP#12 and in 3.17.11.

Failure to complete the test, inability to function and operate as specified or nonconformance to DOT regulations, the California Vehicle Code, to 3.16.5 through 3.16.5.3 or 3.17.11 through 3.17.11.3 shall constitute failure of this test.

4.6.2.6 Tank capacities. Determine the capacities of both the fuel oil tank and the water supply tank. A capacity for either tank of less than 55 gallons shall constitute failure of this test.

4.6.2.7 Lifting attachments. Apply a static load to the lifting attachments as specified in 3.26. Lift the heater and hold in its normal travel position using slings that converge not more than 19 feet above the lowest angles of application for each attachment. Measure the clearance between each sling and the heater.

4.6.2.7.1 Failure criteria. Weld failure or permanent deformation of the heater or its lifting attachments, clearance of less than 1 inch between the slings and the heater, or slings converging more than 19 feet above the lowest extremity of the heater, shall constitute failure of this test.

4.6.2.8 Tiedown attachments. With the heater anchored by the tiedowns, apply a static load as specified in 3.27.

4.6.2.8.1 Failure criteria. Weld failure or permanent deformation of the heater or its tiedown attachments or damage to any tiedown cable shall constitute failure of this test.

4.6.2.9 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.24. The supplier shall furnish the contracting officer the report of test required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

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4.6.2.10 Commercial engine. During the heater test specified in 4.6.2.3 t engine speed (rpm) and intake manifold vacuum shall be measured and recorded at 20 minute intervals for the duration of the test. These readings shall be compared with the engine speed, brake-horsepower and intake manifold vacuum curve for the continuous brake horsepower condition as established during test method 2300, MIL-STD-1400 (see 4.3.1). Any intake manifold vacuum reading recorded during the heater test (4.6.2.3) that is lower than the reading listed on the continuous horsepower curve established during conduct of test method 2300, MIL-STD-1400, at the corresponding speed (rpm), shall constitute failure of this test.

4.6.2.11 Operation. Test and report each heater as specified in 4.6.2.1 and operate each heater to prove functioning of all components. Failure of a boiler or any appurtenance to meet the requirements of any portion of the code, failure of a boiler to be stamped after the test with ASME symbol, or failure of any component to function as required shall constitute failure of this test.

4.7 Inspection comparison. The Government may select heaters at any time during the contract production period and subject these heaters to the examination specified in 4.6.1 and to the tests specified in 4.6.2.1 through 4.6.2.8 to determine that the quality of the selected heater is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Heaters will be selected at random from those which have been e epted by the Government and will not include the previously inspected preproduction model and initial production heater.

4.7.1 Inspection failure. Failure of any heater when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of heaters until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested to determine compliance with the quality assurance provisions of MIL-B-3180.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Each complete heater with repair parts, tools, and technical publications shall be preserved, packaged, packed and marked in accordance with MIL-B-3180. The level of preservation and packaging shall be level A, B, or C and the level of packing shall be level A, B, or C as specified.

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6. NOTES

6.1 Intended use. The bitumen heater is intended for use in heating asphaltic materials shipped in railroad tank cars and automotive type bulk transport tankers during asphalt-plant and surface treatment operation.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2.1).
(c) When initial production inspection is required and number of heaters to be furnished, when applicable (see 3.3).
(d) When a commercial engine is required (see 3.15.2).
(e) When wide-base tires are required (see 3.17.8.2).
(f) When standard military trailer components are required (see 3.19).
(g) When fungus and moisture resistance is not required (see 3.25).
(h) When final paint coloring shall be other than specified (see 3.31).
(i) Repair parts and maintenance tools required (see 3.33).
(k) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production heaters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish heaters conforming to this specification.

6.4 Government-furnished property. When specified, the contracting officer should arrange to furnish the property specified in 3.15.1.

Custodians:

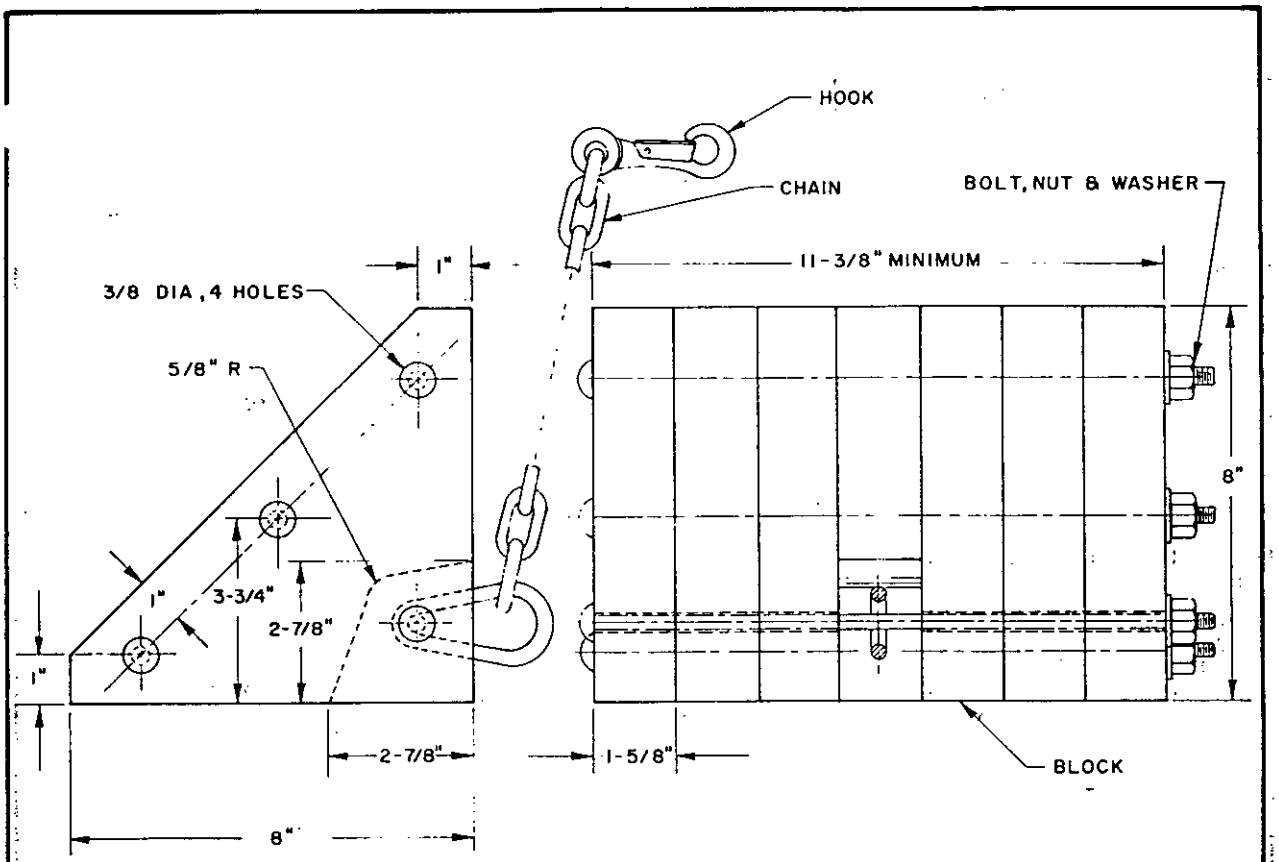
Army - ME
Air Force - 84

Preparing activity:

Army - ME

Project No. 3895-0173

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CHAIN DETAILS

CHAIN WELDED, CLOSE STRAIGHT LINK, GALV'D, 1/4" NOMINAL SIZE OF CHAIN BARS.

LENGTH OF 100 LINKS = 100 INCHES

MINIMUM TENSILE STRENGTH = 3,400 LBS.

WEIGHT/100 FT. = 70 LBS.

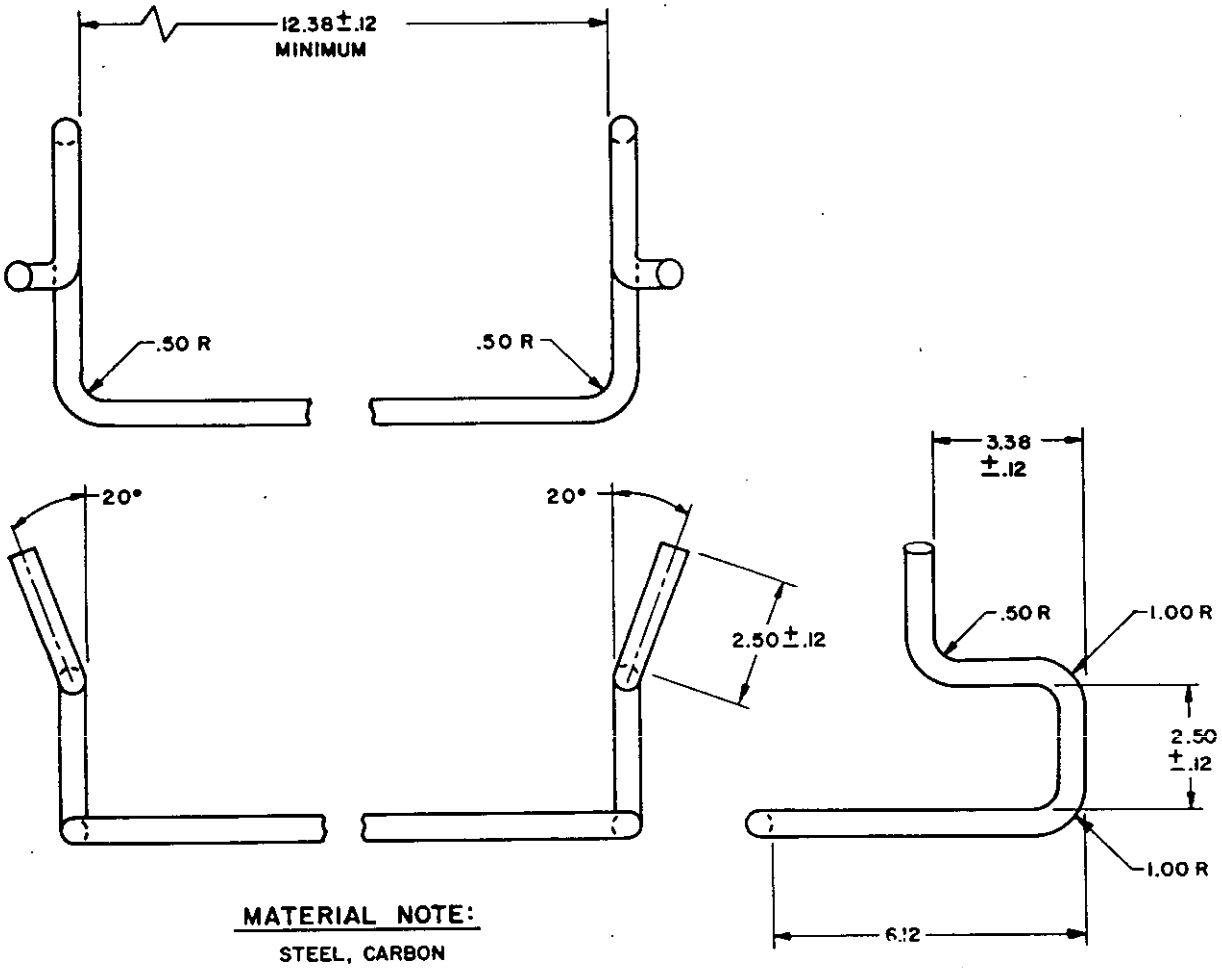
MATERIAL NOTE

WOOD PARTS, FABRICATED, HARDWOOD.

FIGURE 1. CHOCK BLOCK ASSEMBLY

DX-603

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MATERIAL NOTE:

STEEL, CARBON
SAE 1015 TO SAE 1025
1/2 DIA

FIGURE 2. CRADLE

DX-604

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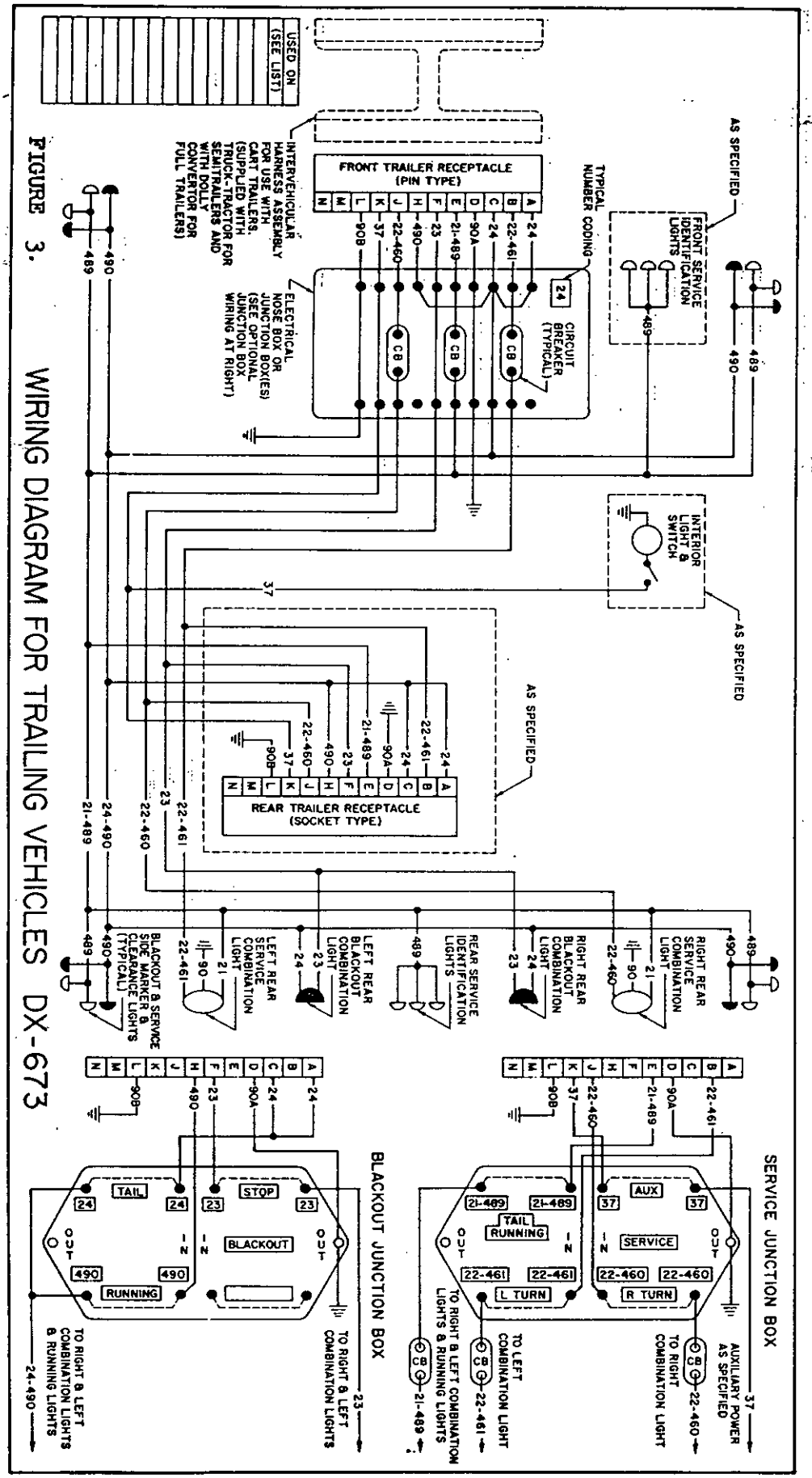


FIGURE 3. WIRING DIAGRAM FOR TRAILING VEHICLES DX-673

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-H-395F- Heater, Bitumen

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE ARMY
U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZG
FORT BELVOIR, VIRGINIA 22060

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RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
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